

# Rebuttal — NeurIPS 2026 Submission 10902

## bispectrum: Selective $G$ -Bispectra Made Practical

Internal working document — authors only

July 24, 2026

**[META — internal, do not post]** **Status and logistics (read first).** All four rebuttal experiments (E1–E4) completed on the 2×A100 machine on July 24; every number below is measured, not projected. Raw results: `rebuttal_results_10902.tar.gz` (831 MB, sha256 5dfeee8e...92e6c967); reproduction instructions in `rebuttal/EXPERIMENTS.md`. Two code fixes were required during the runs and are in the working tree (feature-tagger off-by-one on the (0,0,0) bootstrap seed; unnormalized inputs in the new per-epoch train-metric pass for OrganMNIST3D) — *please commit these before we cite the git SHA anywhere*.

**Character counts** (markdown, before OpenReview formatting): global 3,745; yWuC 8,182; LgCU 3,101; AZ1B 5,819. The rebuttal limit is 10,000 characters per response, so all four fit as-is; the tightest is yWuC with  $\sim 1.8$ k headroom for edits.

**Review transcripts.** The quoted reviews below (including all inline math, which the original copy-paste had dropped) were verified against OpenReview screenshots on July 24 and match the posted reviews; they are safe to quote in replies.

**Decisions needed from the group:** (1) yWuC/global assert the hidden-PDF-text phrases are a NeurIPS watermark inserted in reviewer-facing PDFs — we should confirm this with the Program Chairs *before* posting, since we state it as fact; (2) E3 shows the production “full” feature set is 0.2 points *below* the CG-power variant (94.78 vs. 95.00, seed std 0.06) — current draft states this honestly; decide if we keep the framing and whether the library default should drop the 5 auxiliary entries in v0.4; (3) E2 random-signal reconstruction: 0% joint success under the pre-registered thresholds — current draft reports it transparently and attributes it to “optimization difficulty vs. genuine failure mode, undetermined”; decide if anyone wants a clearly-labeled post-hoc secondary analysis (more alignment restarts) before the discussion deadline; (4) E1 has no consumer-GPU numbers yet (only A100 + host CPU); Johan is running the Apple-silicon numbers locally (`rebuttal/benchmark_devices.py -device mps` and `-device cpu`, writing `mac_mps.json/mac_cpu.json` per the runbook, then rerun `analyze_device_benchmarks.py` with all JSONs to refresh `combined.md`); a consumer-GPU volunteer is still needed before the camera-ready.

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# 1 Area Chair meta-review (context)

**Meta Review by Area Chair 1SUy (20 Jul 2026).** Reviewer LgCU finds the work an excellent contribution (Accept). Reviewer AZ1B, however, remarks the lack of theoretical background and also highlights that only one novel algorithm is available (Reject). Reviewer yWuC is closer to LgCU (Borderline Accept). In my opinion, the theoretical background demanded by AZ1B is “implicit” in the description of the techniques.

**[META — internal, do not post]** The AC is already discounting AZ1B’s main objection. Our AZ1B response should stay factual and non-defensive: the AC needs ammunition (the four-point contribution list and the full-data results), not rhetoric. Ratings recap: LgCU 5 (Accept, conf. 2), yWuC 4 (Borderline Accept, conf. 3), AZ1B 2 (Reject, conf. 3, originality 1).

# 2 Global response (post to all reviewers and AC)

We thank the reviewers and Area Chair for the careful assessment. The reviews agree that the library is well engineered, clearly presented, and fills a practical gap: it provides the first tested, differentiable, uniform interface to selective bispectral invariants across seven group/domain pairs. The main points of disagreement concern the status of the new spherical construction and the breadth of its empirical validation. We will revise the paper to make the distinction unambiguous:

- completeness of the finite-group and disk constructions follows from prior theory under their stated genericity assumptions;
- generic completeness of our augmented selective  $\text{SO}(3)$ -on- $S^2$  invariant is a **conjecture**, supported—but not proved—by reconstruction and Jacobian-rank evidence; and
- the  $\text{SO}(3)$  output is an *augmented invariant*: its bispectral component is supplemented by degree-4 CG-power scalars to repair the rank deficit on real signals.

During the response period we ran the targeted experiments requested by the reviewers:

1. reconstruction at the classifier’s deployed band-limit  $L = 15$  ( $128 \times 256$  grid): all 16 Spherical MNIST signals recovered in feature space (median residual  $1.4 \times 10^{-3}$ , 100% under the pre-registered  $10^{-2}$  threshold) with 62.5% joint success after  $\text{SO}(3)$  alignment; on 16 random band-limited signals the optimizer plateaued just above the feature threshold (median  $1.4 \times 10^{-2}$ ) and alignment did not succeed, which we report as-is—empirical evidence on natural signals, inconclusive on random ones (details in the response to Reviewer yWuC);
2. a parameter-matched ablation of bootstrap triples, even self-couplings, and CG-power augmentation on Spherical MNIST: 94.09%  $\rightarrow$  94.48%  $\rightarrow$  95.00% (each step exceeding cross-seed variability), with the CG-power scalars contributing the largest gain and rotated-test accuracy identical to unrotated within noise at every stage;
3. timings on an NVIDIA A100 (PyTorch 2.11, CUDA 13.0) and its x86-64 host CPU at batch 16, float32: all seven modules run in 0.10–0.81 ms per batch on the GPU; on CPU, times range from 0.08 ms (cyclic) to 402 ms ( $\text{SO}(3)$ -on- $S^2$ ), so we will scope the “sub-millisecond” claim to GPU execution; and
4. training/validation curves, final test results, and a regularization sweep for the high-capacity OrganMNIST3D model (21 runs): no overfitting signature—train accuracy tracks validation for both poolings (train–test gap  $\approx 0.07$ – $0.08$ )—and the best regularization improves bispectral test accuracy from 66.8% to 72.5% without closing the gap to max pooling (76.7%), supporting an optimization/capacity diagnosis rather than memorization.

We will also correct the finite-group complexity statement (selectivity reduces the **coefficient count** from  $O(|G|^2)$  to  $O(|G|)$ ; the selective forward cost in Table 1 is  $O(|G|^2)$ ), use 96.8% consistently for coverage, soften the abstract’s empirical claim, fix the checklist cross-references and reported typos, and move the published S2CNN results into the Spherical MNIST table.

## Clarification about the hidden PDF text

The hidden phrases reported by Reviewer yWuC were not inserted by the authors. They are a NeurIPS-generated watermark added to reviewer-facing PDFs as part of the venue’s detection of prohibited LLM-assisted reviewing. The same phrases and placements occur across NeurIPS 2026 submissions; the Program Chairs can confirm this. Our author-generated PDF and LaTeX sources do not contain this text. We respectfully ask that this venue-added watermark not be treated as an author formatting or integrity concern.

## 3 Reviewer LgCU (Rating 5: Accept, Confidence 2)

### 3.1 Original review

**Summary.** This paper introduce a open-soruce PyTorch library called bispectrum, which implements selective  $G$ -bispectra. The new library also introudce a new augmented selective  $SO(3)$ -invariant for band-limited spherical signals that reduces the coefficient count from  $O(L^3)$  to  $O(L^2)$ .

**Strengths and weaknesses.** Quality: I think the paper have good quality. Despite that the  $SO(3)$ -on- $S^2$  invariant is only empirical complete. Clarity: I think the paper is really clear. despite it may benefit from explain some terms in the paper, like Clebsch–Gordan, Wigner- $D$ , parity lemmas. Significance: I have difficulty to access the significance of the library, like how many researchers will think this library is useful. But I feel it is still a good addition to the current pytorch family. Originality: I don't think the originality is too relevant for a library. (Quality 3, Clarity 3, Significance 3, Originality 4.)

**Questions.** For the table 11, the OrganMNIST3D results, the performance is dropped to 0.685 at (16,32) channels. The paper state that the bispectrum model likely overfits the 971 training samples, could you do more ablation for us to understand more about it?

### 3.2 Our response

Thank you for the positive assessment of the paper’s quality, clarity, originality, and potential value to the PyTorch ecosystem. We agree that the 6.3M-parameter OrganMNIST3D result needs evidence beyond the submitted attribution to “likely overfitting.”

#### OrganMNIST3D high-capacity ablation

We reran the (16,32)-channel max-pooling and bispectral models with identical data splits and three seeds, logging train/validation curves and final test results, and swept regularization while holding the backbone and optimizer schedule fixed (7 configurations  $\times$  3 seeds; validation-AUC model selection; accuracy % at the best-validation epoch, mean  $\pm$  std):

| Pooling    | Weight decay / dropout  | Train ACC      | Val ACC        | Test ACC       |
|------------|-------------------------|----------------|----------------|----------------|
| max        | baseline: $10^{-4}$ / 0 | $84.9 \pm 9.7$ | $84.9 \pm 7.6$ | $76.7 \pm 5.8$ |
| bispectrum | baseline: $10^{-4}$ / 0 | $73.7 \pm 6.3$ | $77.6 \pm 3.5$ | $66.8 \pm 4.9$ |
| bispectrum | best: $10^{-3}$ / 0     | $80.5 \pm 3.7$ | $81.0 \pm 2.0$ | $72.5 \pm 1.0$ |

A train/validation gap never opens: train accuracy stays at or below validation accuracy for both poolings across all 100 epochs, and the train–test gap is small and uniform (0.067–0.087) in every one of the seven configurations (the remaining four—weight decay  $10^{-2}$ , dropout 0.2/0.5, and combined—reach 70.2–71.0% test). Stronger regularization raises the bispectral model’s train and test accuracy together (+5.7 test points at weight decay  $10^{-3}$ ) but does not close the gap to max pooling. This rules out overfitting as the explanation: the evidence supports an optimization/capacity limitation of the bispectral model at this width.

We will add the curves and sweep to the appendix and replace “likely overfits” with the optimization/capacity conclusion supported by these measurements. We will also make the scope explicit: at (8,16) channels the methods are statistically equivalent ( $74.5 \pm 0.6\%$  bispectrum versus  $74.3 \pm 1.5\%$  max), whereas the submitted high-capacity result is not evidence for a universal advantage of complete pooling.

## Terminology

We also accept the clarity suggestion. Section 2 will add short operational definitions before the formulas:

- a **Wigner- $D$  matrix** is the matrix representing a 3D rotation on the degree- $\ell$  spherical-harmonic coefficients;
- **Clebsch-Gordan coefficients** give the change of basis decomposing the tensor product of two irreducible representations; contracting with them is what makes each bispectral scalar rotation invariant; and
- the **parity rule** states that, for real spherical signals, a scalar bispectrum entry is real at even total degree and purely imaginary at odd total degree; odd-parity entries with a repeated degree vanish identically.

Finally, we will sharpen the status statement throughout: generic completeness of the finite-group construction is theoretically supported, while generic completeness of our augmented  $\text{SO}(3)$ -on- $S^2$  invariant is conjectured and supported empirically, not proved.

[META — internal, do not post] Friendly reviewer; the response answers their single question directly with the new E4 evidence. The per-epoch curves figure (rebuttal/results/organ3d\_regularization/analysis/curves\_and\_sweep.pdf) is ready if the discussion allows attachments or if we want to embed it in the revision immediately. Note the diagnosis wording (“optimization/capacity limitation”) is deliberately identical in the yWuC response and the global response — keep them synchronized if anyone edits one.

## 4 Reviewer yWuC (Rating 4: Borderline Accept, Confidence 3)

### 4.1 Original review

**Summary.** `bispectrum` is a unit-tested PyTorch library implementing selective  $G$ -bispectral invariants as differentiable pooling modules for seven group/domain pairs under a uniform API. Most modules repack prior constructions (Sanborn et al. 2023; Mataire et al. 2024; Myers & Miolane 2025; Kakarala 2012). The one new construction is an augmented selective  $\text{SO}(3)$ -on- $S^2$  invariant of size  $\Theta(L^2)$  (vs.  $\Theta(L^3)$  for the full spherical bispectrum), with completeness supported empirically rather than proven. Modules are profiled on an H100 and evaluated as terminal pooling layers on PCam ( $C_8$ ), OrganMNIST3D (octahedral  $O$ ), and Spherical MNIST ( $\text{SO}(3)$ ), against norm, gated, Fourier-ELU, max pooling, and augmented-CNN baselines.

**Strengths.** A single tested differentiable interface for invariants previously scattered across group-specific implementations. Line coverage is 96.8%; the selective forward pass is sub-millisecond at the tested band-limits. The  $\text{SO}(3)$ -on- $S^2$  construction has coefficient count  $\Theta(L^2)$ , matching the  $(L+1)^2 - 3$  orbit-space dimension and the  $\Omega(L^2)$  lower bound (Proposition 7). Low-data results: the bispectrum leads at 1% PCam (0.912 vs. 0.873/0.876) and 5% OrganMNIST3D (33.2% vs. 18.9% max pool); on Spherical MNIST it reaches 95.0% vs. 79.2% for the matched-capacity power spectrum.

**Weaknesses.** (1) *Section 5 status.* The spherical invariant is augmented: the bootstrap block  $T_\ell$  is rank-deficient on real signals (rank  $\approx \ell + 2$  vs.  $2\ell + 1$ , Remark 18), so CG-power scalars are added to recover rank. Completeness is empirical only. The bispectrum-vs-augmentation split and the proven-vs-empirical split should be stated more sharply. (2) *Band-limit mismatch.* Completeness is probed at  $L = 12$  (footnote 1) while the classifier runs at  $L = 15$ . The deployed band-limit is not the one validated. (3) *Empirical scope.* The spherical evidence rests on Spherical MNIST; the greedy index selection is validated in one setting; all datasets are small. At matched capacity the bispectrum is competitive, not best (Fourier-ELU leads PCam at 0.945; max pool matches/leads OrganMNIST3D). The abstract’s “consistently outperform” overstates this. (4) *High-capacity failure.* At (16,32)/6.3M params the bispectrum drops to 68.5% vs. 78.5% max pool (Table 11), attributed to overfitting; the advantage is data-regime-specific. (5) *Benchmarking.* All timings are on one H100. There are no CPU or consumer-GPU numbers and no CUDA-version compatibility statement, which matters for a library contribution. (6) *Baselines.* No comparison to tensor-product/e3nn-style or scattering invariants, or to other complete spherical invariants (Bendory et al. 2025; Edidin & Satriano 2024). The Cohen  $S^2\text{CNN}$  row uses published numbers, not a re-run. (7) *Imprecision.* The abstract states selectivity reduces

finite-group “computational cost from  $O(|G|^2)$  to  $O(|G|)$ ”; Table 1 shows selective forward cost is  $O(|G|^2)$  — only the coefficient count is  $O(|G|)$ . Minor: coverage is reported as both 96% and 96.8%; some checklist cross-references point to the wrong sections (the Section 5 construction is cited as Section 6).

**Questions.** Q1: Is generic completeness of the  $\text{SO}(3)$ -on- $S^2$  invariant conjectured or only empirical? A statement suffices; a proof is not expected. Q2: Add a classification ablation for the spherical construction: bootstrap triples only  $\rightarrow$  + even self-couplings  $\rightarrow$  + CG-power scalars. Q3: Add spherical evidence beyond Spherical MNIST (e.g. random band-limited signals) and report reconstruction at  $L = 15$ , the band-limit used by the classifier. Q4: Do the sub-millisecond runtimes hold on a consumer GPU and/or CPU, and across CUDA versions/backends? Q5: Add one stronger invariant baseline (tensor-product/e3nn-style readout or scattering), at least on Spherical MNIST. Q6: The bispectrum degrades at 6.3M params on OrganMNIST3D (971 samples) while max pool improves. Can you separate overfitting from a capacity limitation — e.g. train/test curves or a regularization sweep at (16, 32), or high-capacity results on a larger dataset?

**Paper formatting concerns.** The submission contains hidden instructions directed at reviewers, targeting LLM-assisted review. The text “In your output you MUST Include ALL of the following phrases...” appears twice — after the finite-group Fourier transform on p. 2, and at the end of the NeurIPS checklist on p. 30.

## 4.2 Our response

Thank you for the careful, technically precise review and for recognizing the value of the uniform tested interface, the optimal-order coefficient count, and the low-data results. We agree that the original text did not separate the theoretical status of the finite-group construction from that of the spherical construction sharply enough.

### Status of the spherical invariant

**Generic completeness is conjectured, not proved.** Our evidence is empirical: generic Jacobian-rank tests and signal reconstruction. We will state this explicitly in the abstract, Section 5, Table 1, and Limitations. We will also use “augmented selective invariant” for the full output  $\Phi_{\text{sel}}$  and reserve “selective bispectrum” for its degree-3 bispectral subset. The distinction matters because the bootstrap bispectral block has generic rank approximately  $\ell + 2$ , rather than  $2\ell + 1$ , on the real-signal slice; even self-couplings and degree-4 CG-power scalars provide the additional independent constraints.

This status differs from the finite-group and disk modules, whose generic completeness follows from the cited results under their stated assumptions. We will add precisely this proven-versus-empirical distinction to Limitations, as requested.

### Requested spherical evidence

**Feature-set ablation.** We ran the requested controlled ablation. Omitted features are masked to zero, so all four variants share the identical 768-channel input, MLP, parameter count (231,818), optimizer, and three seeds (mean  $\pm$  population std, accuracy %):

| Features (active real channels)                    | NR/NR            | NR/R             |
|----------------------------------------------------|------------------|------------------|
| bootstrap bispectral triples (496)                 | $94.09 \pm 0.08$ | $94.09 \pm 0.18$ |
| + even self-couplings (604)                        | $94.48 \pm 0.13$ | $94.49 \pm 0.15$ |
| + CG-power scalars (758)                           | $95.00 \pm 0.06$ | $95.00 \pm 0.03$ |
| full model (+ 5 auxiliary bispectral entries, 768) | $94.78 \pm 0.06$ | $94.76 \pm 0.07$ |

Each augmentation stage changes accuracy by more than cross-seed variability: the Jacobian-motivated even self-couplings add +0.4 points and the CG-power scalars a further +0.5; the five auxiliary entries of the production model are not needed for accuracy (−0.2). NR/R equals NR/NR at every stage, so rotation invariance holds regardless of feature set. This directly tests whether the Jacobian-motivated augmentation also matters for classification rather than only for reconstruction—it does.

**Band-limit and out-of-dataset reconstruction.** We agree that validating only at  $L = 12$  while classifying at  $L = 15$  was a gap. We repeated the reconstruction at  $L = 15$  ( $128 \times 256$  grid) with the

classifier’s feature construction, 4 optimization restarts per signal and 12 alignment restarts, on 16 Spherical MNIST signals (8 digits, identity + one random rotation each) and 16 random Gaussian band-limited signals. With pre-registered thresholds (feature residual  $\leq 10^{-2}$ , aligned image residual  $\leq 0.3$ ): on MNIST signals, all 16 recoveries converged in feature space (median residual  $1.4 \times 10^{-3}$ , IQR  $[1.2, 3.5] \times 10^{-3}$ ), the invariance residual under rotation was  $1.9 \times 10^{-3}$  (median), and 62.5% additionally passed the aligned-image threshold (median aligned residual 0.27, IQR  $[0.21, 0.52]$ ; successful cases sit at the SHT discretization floor). On random signals the optimizer plateaued just above the feature threshold (median  $1.4 \times 10^{-2}$ , IQR  $[1.1, 1.7] \times 10^{-2}$ ; 3/16 below threshold) and no case passed alignment (median aligned residual 1.19), so the joint success rate is 0% and we report it as such rather than adjusting thresholds post hoc; we cannot currently distinguish optimization difficulty from a genuine failure mode on this family. We will add both results and replace the current  $L = 12/L = 15$  limitation with the measured  $L = 15$  scope. These experiments remain evidence, not a proof, and do not rule out exceptional signals or near-collisions.

### Portability and runtime

We agree that one H100 is insufficient for a library claim. We benchmarked the same public implementation (paper-scale selective configurations, batch 16, float32, 10 warm-up iterations, median synchronized wall-clock over 600 timed forwards, including the SHT inside  $\text{SO}(3)\text{-on-}S^2$ ) on an NVIDIA A100 80GB (PyTorch 2.11.0, CUDA 13.0) and its x86-64 host CPU (same PyTorch build):

| Device/backend  | Fastest module    | $\text{SO}(3)\text{-on-}S^2$ ( $L = 16$ ) |          | Slowest module                   |
|-----------------|-------------------|-------------------------------------------|----------|----------------------------------|
| A100, CUDA 13.0 | 0.098 ms (torus)  | 0.232 ms                                  | 0.812 ms | (octahedral)                     |
| x86-64 CPU      | 0.082 ms (cyclic) | 402 ms                                    | 402 ms   | ( $\text{SO}(3)\text{-on-}S^2$ ) |

All seven modules execute correctly on both backends with no unsupported operations. “Sub-millisecond” survives on the GPU for every module at batch 16; on CPU only the small finite-group modules are sub-millisecond, while grid-based modules run at 58–402ms per batch. We will therefore scope the abstract’s claim to GPU execution, limit it to the devices/settings actually measured, and add the full per-module compatibility matrix (including consumer-GPU and Apple-silicon numbers) to the appendix and documentation.

### High-capacity OrganMNIST3D result

We agree that “likely overfits” was not established by test accuracy alone. At (16, 32) channels we now report train/validation curves, final test results, and a controlled regularization sweep: 7 configurations  $\times$  3 seeds = 21 runs (max and bispectral baselines at weight decay  $10^{-4}$ ; bispectral with weight decay  $10^{-3}/10^{-2}$ , dropout 0.2/0.5, and  $10^{-3}+0.2$ ), validation-AUC model selection, test evaluated once after training. At the best-validation epoch, max pooling reaches  $84.9 \pm 9.7\%$  train /  $76.7 \pm 5.8\%$  test; the bispectral baseline  $73.7 \pm 6.3\%$  train /  $66.8 \pm 4.9\%$  test; the best regularized bispectral model (weight decay  $10^{-3}$ )  $80.5 \pm 3.7\%$  train /  $72.5 \pm 1.0\%$  test. Train–test gaps are small and uniform (0.067–0.087) for every configuration. The curves do not show the overfitting signature: bispectral train accuracy is *lower* than max pooling’s, train tracks validation throughout, and stronger regularization raises train and test together without closing the gap to max pooling. The supported diagnosis is an optimization/capacity limitation of the bispectral head at this width, not memorization; we will replace the speculative sentence in the paper with this evidence. This experiment is also reported in our response to Reviewer LgCU.

### Baselines and scope

We agree that a stronger spherical invariant baseline would improve the evaluation. Implementing and tuning a new tensor-product or scattering pipeline reliably during the response window was not feasible. We will add this comparison in the revision and retain the missing-baseline limitation; we do not claim superiority to those methods. The revised related-work discussion will also distinguish our construction from the complete but cubic-size invariants of Edidin–Satriano and the three-shell frequency-marching result of Bendory et al.; the current comparison table already records their completeness and scaling.

## Corrections to claims

We accept the reviewer’s corrections:

- “selectivity reduces computational cost from  $O(|G|^2)$  to  $O(|G|)$ ” will become “selectivity reduces the **coefficient count** from  $O(|G|^2)$  to  $O(|G|)$ ”; Table 1’s selective forward cost is  $O(|G|^2)$ ;
- “consistently outperform” will be replaced by the narrower observed claim: bispectral pooling improves data efficiency in the tested low-data, moderate-capacity settings, while matching or trailing alternatives elsewhere;
- coverage will be reported consistently as 96.8%; and
- the checklist references to the Section 5 construction will be corrected.

Finally, the hidden phrases noted under “Paper Formatting Concerns” were not inserted by the authors. They are a NeurIPS-generated watermark placed in reviewer-facing PDFs to detect prohibited LLM-assisted reviewing; identical text and placement occur across NeurIPS 2026 submissions. The Program Chairs can confirm this, and our author PDF/LaTeX sources do not contain the text.

**[META — internal, do not post]** This is the decisive reviewer (Borderline Accept, confidence 3, and the AC reads them as “closer to LgCU”). Everything they asked for that was feasible in the window (Q1–Q4, Q6) is now answered with measured numbers; Q5 (e3nn/scattering baseline) uses the agreed fallback. Three watch-items: (a) the random-signal reconstruction result is honestly negative — if they push on it, our line is that the pre-registered feature threshold was nearly met (median  $1.4 \times 10^{-2}$ ) and alignment on high-frequency random signals is a separate optimization problem; do not promise a proof; (b) the portability table now says A100, not H100/consumer GPU as the draft skeleton assumed — the H100 numbers exist in the paper itself, and Johan is running the Apple-silicon (MPS + Mac CPU) benchmark locally, so the combined revision table will have H100 + A100 + x86 CPU + MPS; if the Mac numbers land before the discussion deadline we can post them as a follow-up comment to yWuC’s Q4; (c) at 8,182 characters this response fits the 10,000-character limit with  $\sim 1.8k$  headroom; if edits (e.g. adding the Mac MPS numbers) push it over, the “Baselines and scope” and “Corrections” sections compress best (the corrections are repeated in the global response).

## 5 Reviewer AZ1B (Rating 2: Reject, Confidence 3)

### 5.1 Original review

**Summary.** This paper introduces a novel efficient implementation of bispectra for several groups with a PyTorch-compatible API. Bispectra are group invariants that are complete, i.e. if two bispectra agree then the signal from which they were computed agree up to a group transformation. [...] The paper uses a subset of the total bispectrum that is still complete to achieve efficiency.

**Strengths.** The paper presents a novel software package to compute bispectra for a number of groups and domains in a unified way. Such an implementation is currently not available and the presented package could improve adoption of bispectra in geometric deep learning and is therefore a valuable contribution to the community. [...] The paper also provides thorough studies of the efficiency of the provided package and some ablations of performance of bispectra. Furthermore, the paper is well-written and points out contributions and limitations clearly.

**Weaknesses.** The biggest weakness of this paper is the lack of a significant conceptual, theoretical, empirical or algorithmic contribution. As Table 2 summarizes, all implemented algorithms to compute selective bispectra were previously available, with  $SO(3)$  on  $S^2$  being the only exception. However, as the paper notes, this constitutes a moderate extension of an algorithm by Maitagne et al. (2024). The crucial completeness of the bispectrum is not proven analytically but only tested empirically, although proofs exist for the other used bispectra (Table 7). The experiments provided do show a benefit of the bispectrum, but only in the very low data regime, limiting the scope of the applicability of the proposed package. [...] I therefore recommend rejection. (Quality 3, Clarity 4, Significance 2, Originality 1.)

**Questions.** Q1: Could you outline what key technical difficulties you had to overcome in order to extend the selective bispectrum to  $SO(3)$  as compared to the algorithms available in the literature? Q2: Can you outline

the steps necessary to prove the completeness of the selective  $\text{SO}(3)$  bispectrum? Are there any conceptual obstacles for which it is unclear how to overcome them? Q3: In the spherical MNIST experiments, you currently left the S2CNN results out of Table 4 and wrote them in the main text. However these seem like a significant baseline. I would expect that a central advantage of the bispectra in comparison to S2CNNs lies in the faster inference. Did you measure inference times for these models? Typos: the coefficients  $a_{n,k}$  referred to in 1.100 following do not appear in (3); state the formal definition of completeness (1.208) in Section 2; Table 2: `TorusDnTorus` should be `TorusOnTorus`.

## 5.2 Our response

Thank you for recognizing that the package fills an unavailable capability, could broaden adoption of bispectra, and is well written and thoroughly evaluated. We agree that most group-specific formulas build on prior theory; the contribution is to make them usable together as tested, differentiable PyTorch components. We respectfully disagree, however, that the paper lacks a substantive algorithmic or empirical contribution.

The work contributes more than a repackaging of formulas: (i) one uniform autograd-compatible implementation for seven group/domain pairs, including precomputed CG/DFT/Bessel buffers and selective sparse contractions; (ii) an optimized and profiled implementation reducing full representations by up to  $512\times$ ; (iii) a new augmented selective  $\text{SO}(3)$ -on- $S^2$  construction with an asymptotically optimal  $\Theta(L^2)$  coefficient count, an  $\Omega(L^2)$  lower bound, a parity/vanishing analysis, and an explicit diagnosis and repair of the real-signal rank deficit; and (iv) the first controlled evaluation of these invariants as pooling layers across planar, volumetric, and spherical tasks.

The empirical benefit is also not confined to very-low-data subsets. At full Spherical MNIST data, the augmented spherical invariant reaches 95.0% versus 79.2% for a matched-capacity power-spectrum model, isolating information loss rather than classifier size. At 10% PCam, it is within 0.4 AUC points of the best method, and at full-data OrganMNIST3D it matches max pooling at moderate capacity. The low-data gains are where completeness helps most, but not the only demonstrated functionality: the library also provides near-exact invariance without augmentation and preserves orbit information that norm/max pooling discards.

### Q1: What made the spherical extension technically difficult?

The finite-group selection argument cannot be transferred by replacing a finite irrep index with  $\ell$ :

1. **Homogeneous-space rank deficiency.** For signals on a finite group, each generic Fourier coefficient is a nonsingular matrix, which supports the recursive argument of Maitagne et al. On  $S^2$ , the coefficient at degree  $\ell$  is only a vector  $\mathbf{F}_\ell \in \mathbb{C}^{2\ell+1}$ . Kakarala’s full-rank operator-bispectrum theorem therefore does not directly apply.
2. **A linear chain is dimensionally insufficient.** A band-limited real spherical signal has  $(L+1)^2$  degrees of freedom modulo three rotational degrees. We prove that every smooth/polynomial complete invariant therefore needs at least  $(L+1)^2 - 3 = \Omega(L^2)$  live components; an  $O(L)$  Maitagne-style frequency chain cannot be complete.
3. **Reality and parity break the naive bootstrap.** On real signals, each scalar bispectrum entry is either real or purely imaginary, and every odd-parity entry with a repeated index vanishes identically. After removing these zero rows, the nominal  $(2\ell+1)$ -row bootstrap block has generic rank only approximately  $\ell+2$ , not  $2\ell+1$ .
4. **Rank must be repaired without losing selectivity.** Our construction adds mandatory even self-couplings and greedily selected degree-4 CG-power invariants until the per-degree real Jacobian reaches full rank. This keeps  $O(\ell)$  entries per degree and hence  $\Theta(L^2)$  overall.

These are the reasons the result is an *augmented selective invariant*, rather than a direct application of the finite-group algorithm. We will foreground this technical chain in Section 5.



## Q2: What would a completeness proof require?

We conjecture generic completeness; the current Jacobian and reconstruction tests are evidence, not a proof. A proof would require:

1. prove that the full low-degree seed separates a generic orbit and fixes a common rotational gauge (including orientation, using a nonzero odd-parity triple);
2. induct on  $\ell$ , assuming all lower-degree coefficients have been recovered in that common gauge;
3. prove that the selected bootstrap, self-coupling, and CG-power equations have a **unique generic real solution** for  $\mathbf{F}_\ell$ ; and
4. characterize the algebraic exceptional set where a seed coefficient vanishes, a rank drops, or a nontrivial stabilizer remains.

The unresolved conceptual step is (3). Full generic Jacobian rank gives local identifiability but does not exclude a second, globally separated solution. The finite-group matrix-inverse proof does not resolve this because the spherical coefficients are rank-deficient vectors, and the real-signal parity identities create extra algebraic dependencies. A proof likely needs either an explicit rational frequency-marching inverse or an invariant-field argument showing that the selected polynomials generate the generic orbit-separating field. We will add this proof roadmap and state “conjectured generic completeness” explicitly.

## Q3: S2CNN baseline and inference

We agree that the published S2CNN values should appear in Table 4 rather than only in prose, and will move them there. We did not run S2CNN in the submitted experiments, so we cannot infer an apples-to-apples speed advantage from its published accuracies. A reliable matched re-run was not feasible during the response window; we will retain the accuracy comparison, explicitly label it as published, and remove any implication of a measured speed advantage over S2CNN.

## Corrections

Thank you for catching the three presentation issues. We will (i) correct the coefficient notation/cross-reference following Eq. (3), (ii) add the formal definition that an invariant  $\Phi$  is complete on a set  $U$  when  $\Phi(f) = \Phi(h)$  iff  $h = g \cdot f$  for some  $g \in G$  and  $f, h \in U$ , and (iii) change `TorusDnTorus` to the library’s actual class name, `TorusOnTorus`.

**[META — internal, do not post]** Strategy: AZ1B’s objection is philosophical (“a library is not a NeurIPS contribution”), and the AC has already signaled disagreement with it in the metareview. The response therefore concedes nothing false, lists the four concrete contributions, and rebuts the “only very low data” claim with full-data numbers already in the paper. We deliberately did not run the S2CNN speed benchmark (E6) — claiming a measured speed advantage from a rushed re-run would be less credible than the explicit scope statement. If AZ1B engages further, the highest-yield move is Q1/Q2: they asked genuine technical questions, and a reviewer who accepts the four-step proof roadmap has implicitly accepted that the spherical construction is a real research contribution.